

Traffic-Estimate

Hourly traffic forecasting on a SUMO road network (Tehran, District 5 area, 3,229 car edges). Synthetic 24-hour travel demand is simulated with SUMO's mesoscopic engine over many days, and a neural network learns each edge's typical daily traffic profile - a Google-Maps-style "typical traffic" estimator.

Method

1. Zoning - a 3x3 grid of traffic analysis zones (TAZ) over the car edges of the network's largest strongly-connected component (scripts/build_taz.py).
2. Demand - 60 synthetic days as SUMO tazRelation OD matrices, ~6,000 trips/day: gravity-weighted zone pairs, commute direction into the central zone at 07-09 and out at 16-19, peaks at 07:00, 13:00 and 17-19:00, Poisson day-to-day noise (scripts/generate_od.py).
3. Simulation - od2trips -> duarouter -> sumo --mesosim, 24 h per day, ~3.5 s per day; hourly per-edge vehicle counts via edgeData (scripts/run_sim.py).
4. Dataset - 4.6 M rows of (day, edge, hour, vehicles entered) plus static edge features: length, speed limit, lanes, priority (scripts/build_dataset.py).
5. Model - a 64-d learned embedding per edge, an hour embedding with sin/cos time encoding and the static features feed a 2x256 MLP; a per-edge Fourier head (6 daily harmonics) gives every edge its own smooth 24 h profile. Trained with Poisson negative log-likelihood, the correct loss for counts, on 49 days; 3 validation and 8 test days are held out (scripts/train.py). A GRU profile decoder and a per-(edge,hour) historical-average table are available as alternatives and score the same, so the MLP is at the noise ceiling.

Model usage

Input: edge id (any SUMO edge of the network) + hour of day (0-23; "17:30" is accepted and truncated to the hour).

Output: expected vehicles entering that edge during that hour on a typical day (vehicles/hour).

```
$ python scripts/predict.py --model-dir output/model \
  --edge "330920957#0" --time 7
edge 330920957#0 at 07:00 -> 85.8 vehicles/hour

$ python scripts/predict.py --model-dir output/model \
  --edge "330920957#0" --profile          # full 24h curve

# python API
from predict import load_predictor
meta, fn = load_predictor("output/model")
rates = fn(meta["edges"].index("330920957#0"))[0] # 24 hourly rates
```

Results (held-out simulated days)

metric	single day	typical day
accuracy (within 20% or 3 veh)	95.0%	99.9%
R2	0.950	0.992
MAE (veh/h)	0.82	0.31

"Typical day" scores the prediction against the average of the test days - the quantity a typical-traffic forecast estimates. The ~5% single-day residual is the irreducible Poisson randomness of daily demand: an oracle lookup table scores the same.

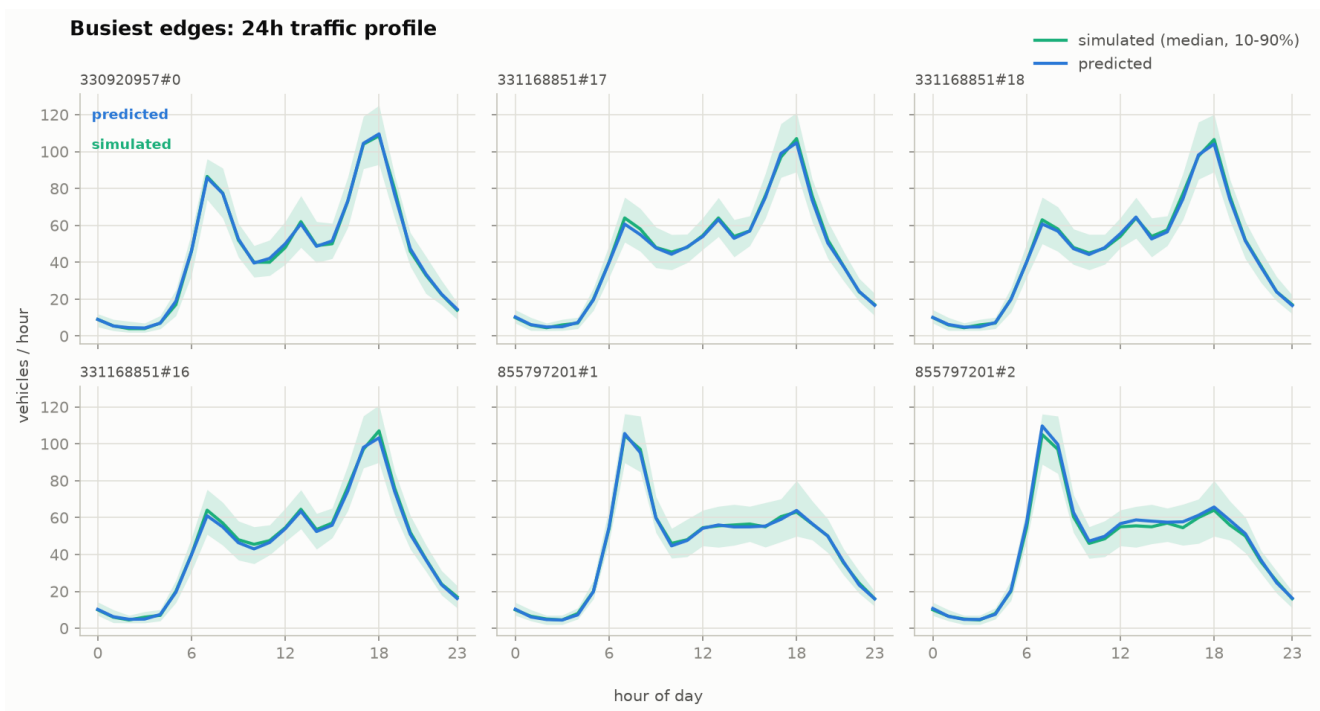
Estimated traffic across the network

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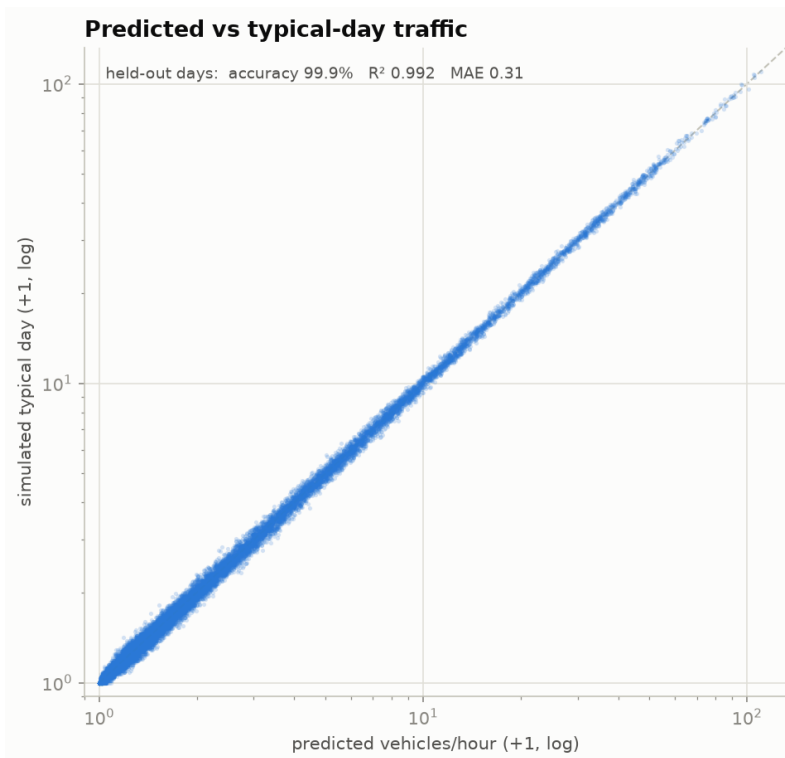


Predicted vehicles/hour on every edge at 07:00, 13:00, 18:00 and 23:00. Arterials carry the load; 18:00 is the heaviest hour.

Model vs simulation

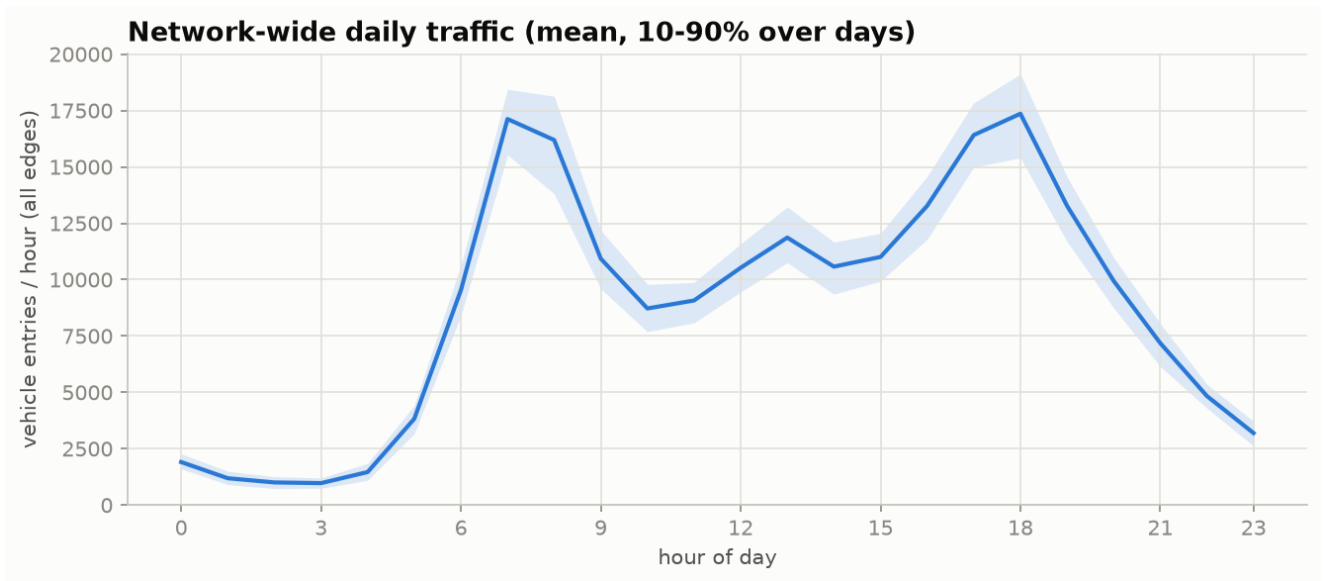


24 h profiles of the six busiest edges: model prediction (blue) vs simulated median and 10-90% band across days (green).



Prediction vs typical-day simulated traffic for every (edge, hour) pair, log scale.

Network-wide daily demand



Total vehicle entries per hour across all edges (mean and 10-90% band over the 60 simulated days): morning peak at 07:00, midday bump at 13:00, evening peak at 18:00.